QA Review Report

Project 002 – Sentiment Analysis

---

1. Overview

This report summarizes the Quality Assurance (QA) process conducted during the sentiment annotation of 150 customer feedback messages. The review process was designed to improve annotation consistency, resolve ambiguous cases, and ensure that all labels aligned with the project's annotation guidelines. Rather than validating annotations against a predefined answer key, the review emphasized reasoning based on evidence and consistent application of annotation principles.

---

2. QA Objectives

The QA process aimed to:

- Verify annotation accuracy and consistency.
- Identify incorrect or inconsistent sentiment labels.
- Review ambiguous customer feedback.
- Resolve disagreements through structured adjudication.
- Refine annotation guidelines based on recurring edge cases.
- Produce a finalized dataset suitable for AI training and evaluation.

---

3. Review Methodology

The dataset was reviewed in six batches of twenty-five messages.

For each batch:

1. Initial annotations were completed.
2. Annotation rationale was documented for ambiguous examples.
3. Reviewer comments were provided for challenging cases.
4. Disagreements were discussed and adjudicated.

5. Corrections were applied where necessary.
6. The final approved labels were transferred to the production dataset.

This iterative review process improved consistency throughout the project.

---

4. Common Review Criteria

Annotations were evaluated using the following criteria:

- Overall customer sentiment.
- Explicit versus implied opinion.
- Dominant versus balanced sentiment.
- Factual statements versus evaluative language.
- Expectation fulfilment or violation.
- Product functionality versus secondary characteristics.
- Service recovery effectiveness.
- Consistency with previously reviewed examples.

---

5. Representative QA Findings

Several recurring annotation challenges were identified during review.

- **Distinguishing Neutral from Positive**

Messages confirming successful actions (e.g., payment received, application under review, shipment updates) were carefully distinguished from messages expressing genuine customer satisfaction.

- **Mixed Sentiment**

Messages containing both positive and negative opinions were evaluated to determine whether both perspectives materially contributed to the overall meaning or whether one sentiment clearly dominated.

- **Expectation Violations**

Customer feedback indicating that products or services failed to meet advertised or expected performance consistently resulted in stronger negative sentiment.

Examples included:

- misleading product descriptions
- poor durability
- delayed deliveries
- underperforming product features

- **Service Recovery**

Special attention was given to situations where companies responded effectively after an initial problem. Recovery actions such as prompt apologies, quick replacements, or efficient issue resolution were evaluated to determine whether they outweighed the original negative experience.

- **Core Functionality**

Product reviews required distinguishing between cosmetic features and core functional performance. Negative sentiment was considered dominant when primary functionality failed despite secondary positive characteristics.

---

6. Guideline Refinements

Reviewer calibration resulted in several improvements to the annotation guidelines.

Key refinements included:

- Successful status updates do not automatically indicate Positive sentiment.
- The presence of the word "but" does not automatically require a Mixed label.
- Core functionality may outweigh secondary product attributes.
- Restoration of expected functionality after repeated customer effort does not necessarily constitute Positive sentiment.
- Annotation decisions should remain evidence-based and avoid unsupported assumptions.

These refinements improved consistency across later review rounds.

---

7. Quality Outcomes

The QA process resulted in:

- Improved annotation consistency across all six review batches.
- Reduced disagreement rates as reviewer calibration progressed.
- Consistent handling of edge cases.
- A finalized dataset supported by documented reviewer feedback and adjudication.

The completed dataset reflects both initial annotation and structured quality assurance.

In conclusion, the review process demonstrated that high-quality sentiment annotation depends on more than identifying positive or negative words. It requires:

- contextual interpretation,
- consistent guideline application,
- structured reviewer calibration,
- thoughtful handling of ambiguity, and
- evidence-based decision-making.

Collaborative adjudication played an important role in strengthening both the dataset and the annotation guidelines. The Quality Assurance process significantly improved annotation quality by identifying inconsistencies, resolving ambiguous cases, and refining project guidelines. The resulting dataset represents a realistic sentiment annotation workflow comparable to those used in professional AI data annotation projects and provides a reliable resource for demonstrating practical annotation and QA skills.